



ಭಾರತೀಯ ಮಾಹಿತಿ ತಂತ್ರಜ್ಞಾನ ಸಂಸ್ಥೆ ರಾಯಚೂರು
भारतीय सूचना प्रौद्योगिकी संस्थान रायचूर
Indian Institute of Information Technology Raichur

DSA : LAB - 6

M&C Batch 2024-2028

Course Instructor: Dr. Chandra Mohan D

Total Marks: 100

Time Duration: 2 Hours

Instructions (Read Carefully)

- **No AI Assistance:** Use of any AI tools for code generation is strictly forbidden.
- **Direct Code Entry Only:** Do not prepare code on paper or external sources; all work must be typed live.
- **Zero Tolerance for Cheating:** Any form of cheating will result in a **zero score for this exam and the past four labs.**
- **Academic Integrity is Mandatory:** Violations will have severe penalties—there will be no exceptions.
- **No Unauthorized Materials:** Only the approved exam tools and resources may be used; personal notes or external references are not allowed.
- **Strict Supervision:** Exam sessions will be closely monitored; any suspicious behavior will be investigated immediately.
- **Immediate Consequences:** Any breach of these rules will lead to immediate and severe disciplinary action.

Additional Rules Regarding TAs and Marks

- **No External Queries to TAs:** Outside of lab sessions, TAs will only respond to classroom comments. Questions or complaints outside the session will not be entertained.
- **Marks Disclosure:** The disclosure of marks is entirely at the discretion of the faculty. Students are not allowed to request mark changes, complain, or pressure TAs in any form once the session is completed.



Question 1: The Writer's Diary (Undo-Redo) {25 Marks}

Story

A writer owns an old mystical typewriter. Each time they press a key, the character magically appears at the end of the diary page. The typewriter, however, has two strange levers:

- One lever can reverse the last thing you did.
- Another lever can bring back an action you had just undone.

The writer gives you a log of everything they did during the day. By the end, they want to know how their diary looks.

But there's a catch:

- If nothing is written and the undo lever is pulled, the typewriter simply makes a click sound (ignore it).
- If nothing remains in the diary at the end, you must report EMPTY.

Problem Statement

Given a sequence of operations (TYPE, UNDO, REDO), determine the final text in the diary.

Input Format

- First line: integer N (number of operations).
- Next N lines: each operation is one of the following:
 - TYPE X → type the character X
 - UNDO → undo the last operation
 - REDO → redo the most recently undone operation

Output Format

Print the final diary content.

If the diary is blank, print EMPTY.

Constraints

- $1 \leq N \leq 10^5$
- X is a single English letter (a–z or A–Z).
- The number of UNDO and REDO operations will always be logically valid.

Sample Test Case

Input:	Output:
4 UNDO TYPE M UNDO REDO	M Execution: UNDO → nothing → EMPTY TYPE M → M UNDO → remove M → EMPTY REDO → restore M → M

6 TYPE A TYPE B TYPE C UNDO REDO TYPE D	ABCD
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Question 2: The Trader's Curse (Stock Span) {35 Marks}

Story

A greedy trader believes he can predict luck in the stock market. His method is odd:

For every day's price, he looks backwards and counts:

- How many consecutive earlier days had a price less than or equal to today's?
- He includes today itself in this count.

If today's price is high, his count grows long. If today's price is small, the count resets to 1.

You are given the list of prices. The trader demands you compute this mysterious number for each day.

Problem Statement

For each day i , compute the number of consecutive days (including day i) for which the stock price was $\leq$ price on day i .

Input Format

- First line: integer $N \rightarrow$ number of days.
- Second line: N integers $\rightarrow$ stock prices.

Output Format

Print N integers separated by spaces — the trader's number for each day.

Constraints

- $1 \leq N \leq 10^5$
- $1 \leq \text{Price} \leq 10^9$

Sample Test Case :

Input:	Output:
6 10 4 5 90 120 80	1 1 2 4 5 1 Execution: Day 1: 10 $\rightarrow$ span 1 Day 2: 4 < 10 $\rightarrow$ span 1 Day 3: 5 > 4 $\rightarrow$ span 2 Day 4: 90 > 5,4,10 $\rightarrow$ span 4

	Day 5: $120 > \text{all before} \rightarrow \text{span } 5$ Day 6: $80 < 120 \rightarrow \text{span } 1$
7 100 80 60 70 60 75 85	1 1 1 2 1 4 6

Question 3: The Divine River of Souls {40 Marks}

Story

The ancient map in your weathered hands has led you through treacherous mountain passes and whispering forests. As the final mist clears, you find yourself standing before something extraordinary—the legendary *River of Souls*. The water doesn't flow normally; it seems to shimmer with divine light and ancient wisdom.

An ethereal voice echoes from the depths: "Welcome, traveler. You stand at the boundary between two realms. This river... it separates the mortal world from the divine."

You notice something profound. On the *Left Bank* - the *Mortal Realm* - countless souls stand waiting with desperate hope in their eyes, clutching carved stones with mystical words. They all yearn to cross to the *Right Bank* - the *Realm of Ascension* - where eternal paradise awaits.

But on the Right Bank, you see a curious sight: some souls already there seem troubled, holding stones with words that lack power. These souls tremble, knowing they may be cast back to mortality.

As you approach the water's edge, the divine voice grows clearer: "This is no ordinary crossing, mortal. The river judges the purity of souls through the words they carry."

The river's surface begins to glow with celestial light, and suddenly you see visions of the divine judgment:

- *Vision 1*: Souls from the Mortal Realm can only ascend if their words possess perfect symmetry - reading the same in both directions of time
- *Vision 2*: Souls already in the Ascension Realm who carry impure words are cast back to mortality
- *Vision 3*: The judgment happens in a divine rhythm, but something about the order seems... backwards from mortal expectations

"These visions show fragments of divine truth. But to truly understand the celestial pattern, you must discover it yourself."

The water begins to swirl with holy light, and angelic figures emerge—the spirits of ancient coders who once faced this same divine trial. "The river will test more than your ability to cross," one angel whispers. "It will test your ability to **manage the flow of souls**, to **understand divine patterns**, and to **create order from spiritual chaos**."

Another spirit adds: "Some souls will find new words of power. Others will abandon their quest. Some will return with renewed hope. The divine river remembers every soul... do you?"

The river itself speaks with the voice of eternity: "I will grant you the power to witness and orchestrate this sacred sorting of souls. But first, you must prove you can understand divine justice through the ancient art of **logical construction**."

Suddenly, the scene becomes alive with spiritual activity. Souls are arriving at the Mortal Realm, desperate to ascend. Some whisper new words of power to their stones. Others abandon hope and vanish. Meanwhile, souls in the Ascension Realm who lack pure words face the terror of being cast back to mortality.

The river shows you the divine laws:

- "Words of perfect symmetry - palindromes - are the keys to ascension"
- "But divine judgment sees beyond mortal limitations of 'case' and 'symbols'"
- "The rhythm of judgment follows the pattern of the last to arrive being first to be judged"

You witness glimpses of the sorting ritual:

- "From mortality to divinity, only the pure may pass"
- "From divinity to mortality, the impure must fall"
- "The dance continues until all souls find their rightful realm"

The divine voice resonates with infinite power: "I sense the spark of a **creator** in you—one who can forge **pathways between realms** and **command the language of divine logic**. Very well."

"I grant you dominion over this sacred sorting, but only if you can create a **divine artifact**—a construction of pure logic that can witness the movements of souls, judge the purity of their words, orchestrate the sacred dance between realms, and speak the final accounting of where each soul belongs."

"Your artifact must be written in the **Sacred Language of C**, for only through such precise divine syntax can the river's celestial magic be properly channeled."

Input Format

The river speaks in the following divine tongue:

N M

[N operations for the Mortal Realm (Left Bank)]

[M operations for the Ascension Realm (Right Bank)]

JUDGE

Where operations include:

- ADD <name> <word> - A soul arrives at the realm
- REMOVE <name> - A soul abandons their quest
- CHANGE <name> <new_word> - A soul discovers a new word of power

Output Format

- *First line*: Names of souls who achieved ascension and remain in the divine realm

- *Second line*: Names of souls condemned to remain in or cast back to the mortal realm

Constraints

- $1 \leq N, M \leq 1000$
- Names are strings of up to 50 characters
- Words are strings of up to 100 characters
- Names contain only letters and may have spaces
- Words may contain letters, numbers, spaces, and punctuation

Sample Test Cases :

Test Cases	Input:	Output:
1	3 2 ADD Alice racecar ADD Bob hello ADD Charlie madam ADD Diana A man a plan a canal Panama ADD Eve world	Charlie Alice Diana Bob Eve
2	2 1 ADD John level REMOVE John	Mary Peter

	ADD Mary kayak ADD Peter noon	
3	2 2 ADD Tom abc CHANGE Tom Taco Cat ADD Sarah xyz ADD Mike deed CHANGE Mike test ADD Lisa stats	Tom Lisa Sarah Mike

Your Divine Quest

Write a complete C program that can commune with the Divine River of Souls. Your code must demonstrate mastery of divine data structures, sacred string processing, and celestial algorithm design while discovering the river's holy patterns through careful observation of the sample cases.

The divine mechanics you must discover:

- Souls from the Mortal Realm ascend to the Divine Realm only if their words are palindromes
- Souls in the Divine Realm are cast back to the Mortal Realm if their words are not palindromes
- The judgment follows the pattern where the last to arrive is first to be judged (divine stack order)
- Divine palindrome detection transcends mortal concerns of case and symbols

The river of souls awaits your divine creation. The souls grow restless between realms. Only through your artifact can the eternal sorting be completed.

May your algorithms be blessed and your pointers guide souls to their rightful destiny. ✨

"In the divine pattern, all souls find their true home..." - *The River of Souls*
